

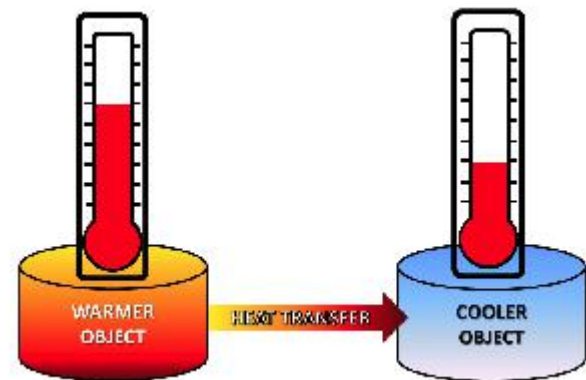
8.1 Heat

8.2 Ideal gas equation

8.3 Thermodynamics



The
Gas
Laws





Learning Outcomes

8.1 HEAT

At the end of this chapter, students should be able to:

- a) Define** heat conduction
- b) Solve** problems related to rate of heat transfer through a cross sectional area (only one material)
- c) Discuss** graph of temperature-distance, T - x for heat conduction through insulated material (one material & lagged material)



8.2 IDEAL GAS EQUATION

At the end of this chapter, students should be able to:

- a) State** Gas's Law
- b) Sketch** the following graphs of an ideal gas :
 - i) p-V graph at constant temperature
 - ii) V-T graph at constant pressure
 - iii) p-T graph at constant volume
- c) Explain** the following graphs of an ideal gas :
 - i) p-V graph at constant temperature
 - ii) V-T graph at constant pressure
 - iii) p-T graph at constant volume
- d) State** ideal gas equation.
- e) Use** ideal gas equation, $pV = nRT$



Learning Outcomes

8.3 THERMODYNAMICS

At the end of this chapter, students should be able to:

- a) State** the first law of thermodynamics.
- b) Solve** problems related to the first law of thermodynamics.
- c) Define** the thermodynamics processes :
 - i) Isothermal
 - ii) Isochoric
 - iii) Isobaric
 - iv) Adiabatic
- d) Interpret** p - V graph for all the thermodynamics processes.
(Experiment 6 : Heat)

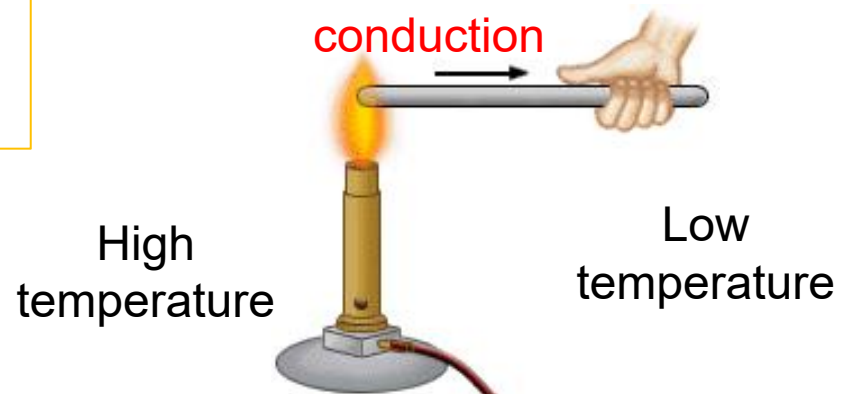
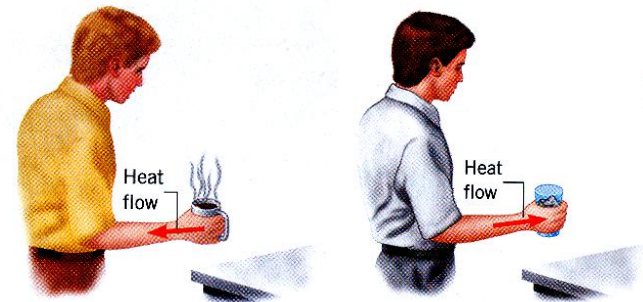
8.1a Heat

- ✓ is the **energy transferred** from one object to another because of a **difference in temperatures** between the two objects.
- ✓ Heat flows from **higher** temperature **to lower** temperature.

Unit of heat is **Joule (J)** , $\text{kg m}^2 \text{s}^{-2}$

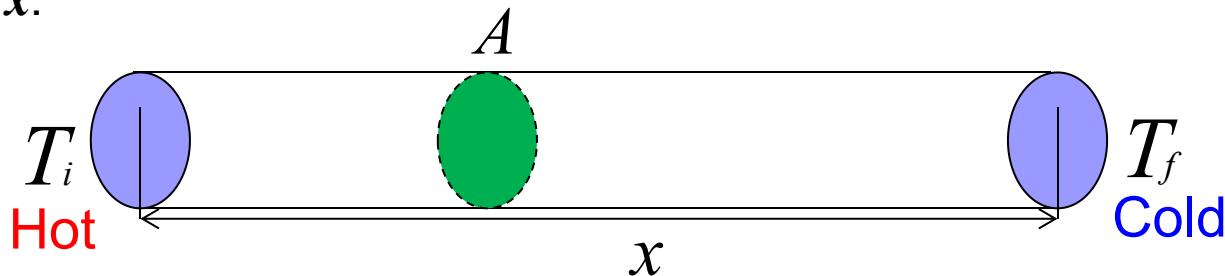
Heat conduction

is defined as a process whereby heat is transferred through a solid from a region of high temperature to a region of lower temperature.



8.1b Rate of heat Conduction

Consider a rod of conducting material with uniform cross sectional area, A & length x .



Experiments show that the rate of heat flow is given by the relation

$$\frac{dQ}{dt} = -kA \left(\frac{dT}{dx} \right)$$

OR

$$\frac{dQ}{dt} = -kA \left(\frac{T_f - T_i}{x} \right)$$

Unit: W @ J s⁻¹

where,

 $\frac{dQ}{dt}$

Rate of heat flow

 $\frac{dT}{x}$

Temperature gradient OR

Temperature difference per unit length

 A :

Area

 k :

Thermal conductivity

 dT :Temperature difference ($dT = T_f - T_i$)

But because dT is negative (-)
So, rate of heat flow will always be positive (+)

Note:

(-) sign in the equation shows that heat always **flow in the direction of decreasing temperature**.

Note:

The temperature change, ΔT is the same in the Kelvin and Celsius scales.

Thermal conductivity, k

- is the **characteristic of heat conducting ability of a material**.
- Indicator of how fast a material able to conduct heat.

Unit for k : $\text{W m}^{-1} \text{K}^{-1}$ @ $\text{W m}^{-1} \text{°C}^{-1}$

scalar quantity

-- is defined as a **rate of heat flows** perpendicularly **through the unit cross sectional area of a solid**, per unit temperature gradient along the direction of heat flow

Substance	Thermal conductivity, k ($\text{W m}^{-1} \text{K}^{-1}$)
Silver	420
Copper	380
Steel	40
Glass	0.84
Brick	0.84
Wood	0.1

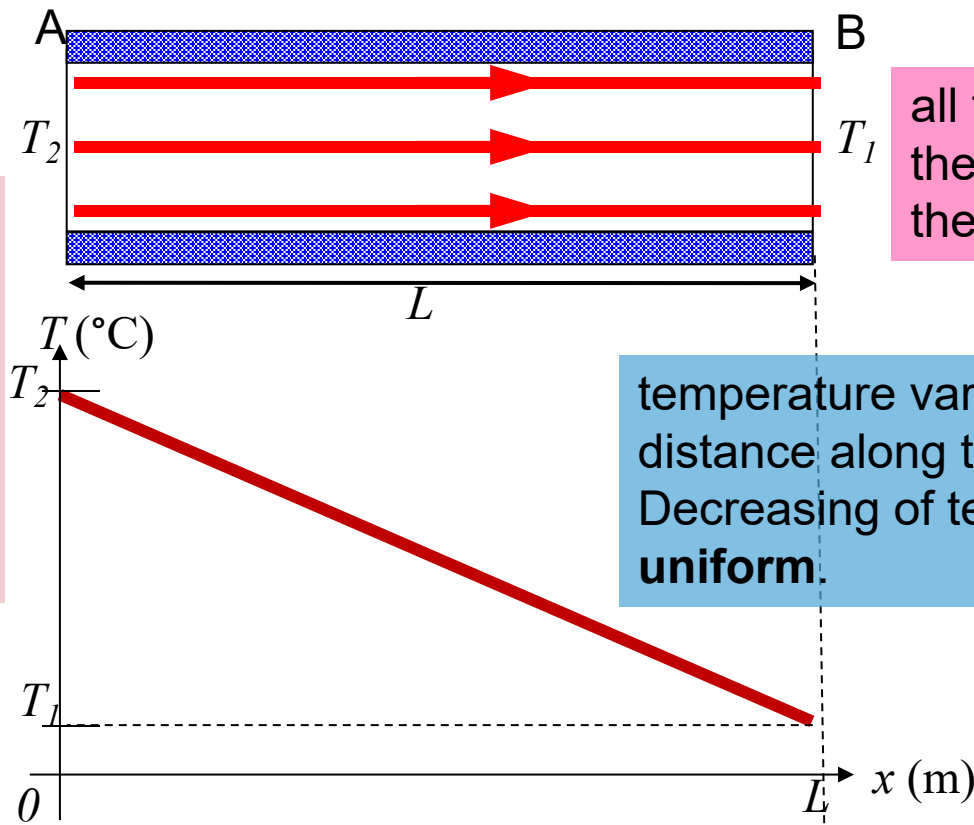
Good conductor

Poor conductor /
Good insulator

8.1c Graph of Temperature-distance, T - x

Conduction along a well Insulated rod

No heat loss through the side surface of the rod because it is **covered with insulator**.

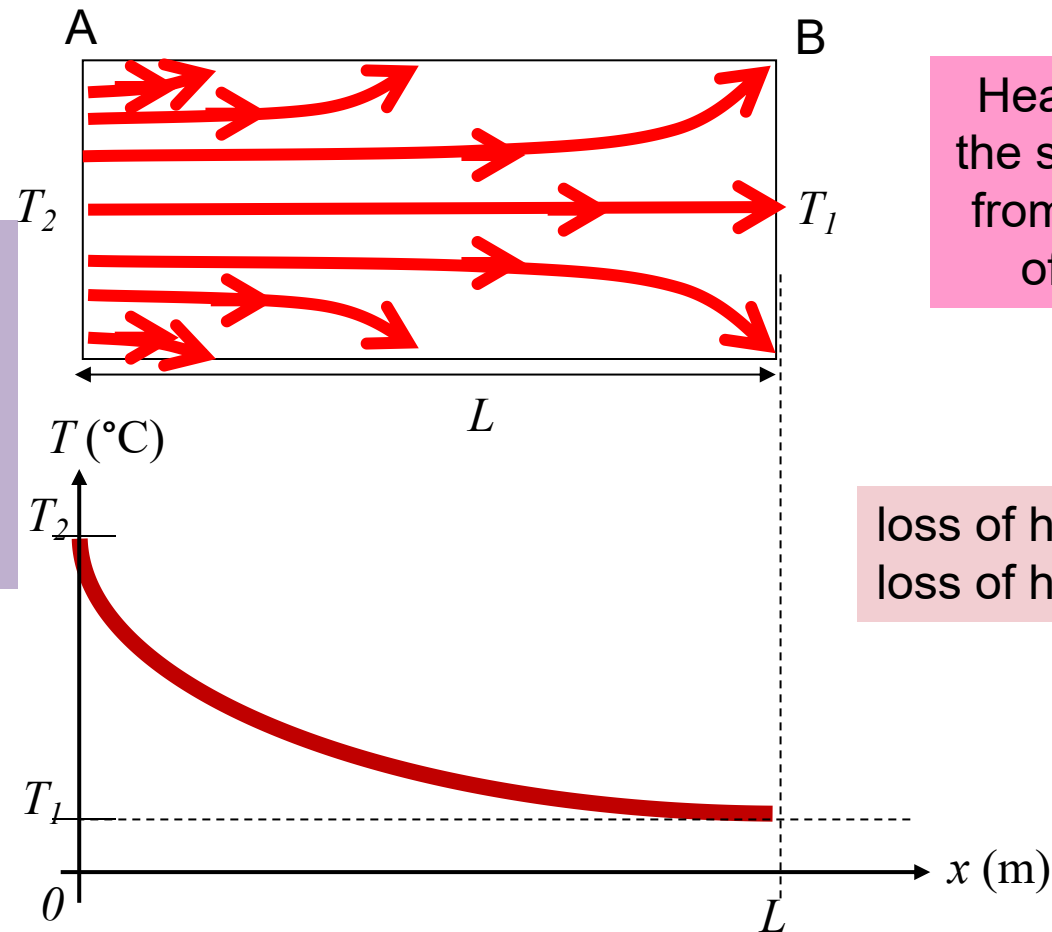


all the heat flow from the hot end of A to the cold end of B.

temperature varies linearly with distance along the rod. Decreasing of temperature is **uniform**.

Conduction along non - insulated rod

temperature – distance graph is a curve. The decreasing of temperature is not uniform.



Heat is lost to the surrounding from the sides of the rod

loss of heat at A > loss of heat at B

Example 1

Calculate the rate of heat flow through a glass window 3.0 m^2 in area and 3.2 mm thick if the temperature at the inner and outer surfaces are 15°C and 38°C respectively.

[given $k_{\text{glass}} = 0.84 \text{ W m}^{-1} \text{ K}^{-1}$]

Solution

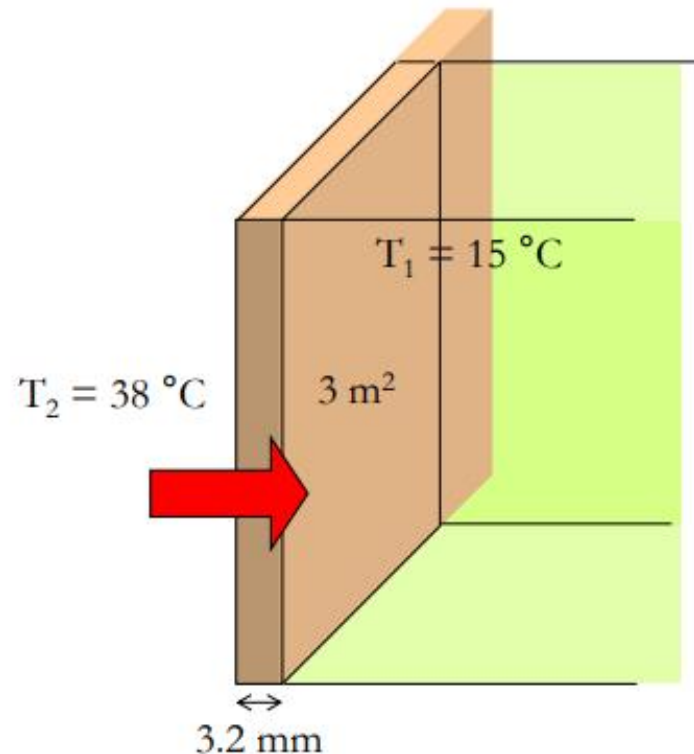
Given: $A = 3.0 \text{ m}^2$; $x = 3.2 \times 10^{-3} \text{ m}$

$T_i = 38^\circ\text{C}$; $T_f = 15^\circ\text{C}$;

$dT = T_f - T_i = 15 - 38 = -23^\circ\text{C}$

$$\begin{aligned}\text{Using : } \frac{dQ}{dt} &= -k A \frac{dT}{x} \\ &= -(0.84)(3) \frac{-23}{3.2 \times 10^{-3}}\end{aligned}$$

$$= 1.81 \times 10^4 \text{ W}$$



Example 2

A Styrofoam box used to keep drinks cold at a picnic has total wall area (including the lid) of 0.8 m^2 and wall thickness 2.0 cm . It is filled with ice, water, and cans of Omni-Cola at 0°C . What is the rate of heat flow into the box if the temperature of the outside wall is 30°C ?

($k_{\text{styrofoam}} = 0.01 \text{ W m}^{-1} \text{ K}^{-1}$)

Solution

Given: $A = 0.8 \text{ m}^2$; $x = 2.0 \times 10^{-2} \text{ m}$

$T_i = 30^\circ\text{C}$; $T_f = 0^\circ\text{C}$;

$dT = T_f - T_i = 0 - 30 = -30^\circ\text{C}$

Using : $\frac{dQ}{dt} = -k A \frac{dT}{x}$

$$= -(0.01) (0.8) \frac{-30}{0.020}$$

$$= 12 \text{ W}$$

Example 3

The temperature on the outer and inner surfaces of a glass window pane are 35°C and 20°C . How fast is heat conducted through the window pane if its surface area is 3000 cm^2 and thickness is 3.0 mm ?

($k_{\text{glass}} = 0.80\text{ W m}^{-1}\text{ K}^{-1}$)

Solution

Given: $A = 0.3\text{ m}^2$; $x = 3.0 \times 10^{-3}\text{ m}$

$T_i = 35^{\circ}\text{C}$; $T_f = 20^{\circ}\text{C}$;

$dT = T_f - T_i = 20 - 35 = -15^{\circ}\text{C}$

$$\begin{aligned}\text{Using : } \frac{dQ}{dt} &= -k A \frac{dT}{x} \\ &= -(0.8)(0.3) \frac{-15}{3 \times 10^{-3}}\end{aligned}$$

$$= 12\,000\text{ W}$$

Exercise 8.1

1. A well insulated aluminium bar has 40 cm length and cross-sectional area 3.0 cm^2 . one of the ends is maintained at a temperature of 130°C while its other end is kept at a temperature of 10°C . If the bar is in steady state condition, find

(a) The temperature gradient along the bar

(b) The rate of heat flow through the bar.

($k_{\text{aluminium}} = 240 \text{ W m}^{-1} \text{ K}^{-1}$)

2. A copper rod has a diameter of 3 cm and thickness of 0.6 m. One end of the rod is placed in boiling water and the other end is in ice. Calculate the quantity of heat transferred through the rod within one minute.

($k_{\text{copper}} = 380 \text{ W m}^{-1} \text{ K}^{-1}$)

3. A refrigerator has walls with thermal conductivity of $0.05 \text{ W m}^{-1} \text{ K}^{-1}$, thickness of 90 mm and an effective area of 5.6 m^2 . The space inside the refrigerator must be maintained at a temperature of 25°C lower than the outside temperature. Calculate the minimum power of the refrigerator.

Exercise 8.1

4. A glass window of cross sectional area 1.5 m^2 and thickness 0.2 cm is closed in winter. The temperature of the inner and outer surfaces of the window are 15°C and 0°C . ($k_{\text{glass}} = 0.84 \text{ W m}^{-1} \text{ K}^{-1}$)

- a) Calculate the rate of heat flow through the window.
- b) Suggest how would you reduce the amount of heat loss to the surroundings through this window.

5. Sketch graphs on the same axes to show the conduction of heat through metal conducting rod when the rod is insulated and not insulated. Explain briefly the shapes of the graphs you have draw.

ANS:

1. a) $-300^\circ\text{C m}^{-1}$ b) 21.6 W 2. 2686.67 J 3. 77.78 W 4. 9450 W

Solution

1. Given: $A = 3 \times 10^{-4} \text{ m}^2$, $x = 40 \times 10^{-2} \text{ m}$, $T_i = 130^\circ\text{C}$, $T_f = 10^\circ\text{C}$, $k = 240 \text{ W m}^{-1} \text{ K}^{-1}$

(a) The temperature gradient along the bar is, $\frac{dT}{dx} = ?$

$$\begin{aligned}\frac{dT}{dx} &= \frac{T_f - T_i}{x} \\ &= \frac{10 - 130}{40 \times 10^{-2}}\end{aligned}$$

$$= -300^\circ\text{C m}^{-1}$$

(b) The rate of heat flow through the bar is, $\frac{dQ}{dt} = ?$

$$\begin{aligned}\frac{dQ}{dt} &= -k A \frac{dT}{dx} \\ &= -(240) (3 \times 10^{-4}) (-300)\end{aligned}$$

$$= 21.6 \text{ W}$$

Solution

2. Given: $d = 0.03 \text{ m}$, $A = \frac{\pi d^2}{4} = 7.07 \times 10^{-4} \text{ m}^2$,
 $x = 0.6 \text{ m}$, $T_i = 100 \text{ }^\circ\text{C}$, $T_f = 0 \text{ }^\circ\text{C}$,
 $k = 380 \text{ W m}^{-1} \text{ K}^{-1}$, $t = 1 \times 60 = 60 \text{ s}$,
 $dQ = ?$

$$\frac{dQ}{dt} = -kA \frac{T_f - T_i}{x}$$

$$dQ = -380 \times 7.07 \times 10^{-4} \times \frac{0 - 100}{0.6} \times 60$$

$$dQ = 2686.6 \text{ J}$$

Solution

3. Given: $A = 5.6 \text{ m}^2$, $x = 90 \times 10^{-3} \text{ m}$,
 $dT_i = -25 \text{ }^\circ\text{C}$, $k = 0.05 \text{ W m}^{-1} \text{ K}^{-1}$, $\frac{dQ}{dt} = ?$

$$\begin{aligned} \frac{dQ}{dt} &= -kA \frac{T_f - T_i}{x} \\ &= -0.05 \times 5.6 \times \frac{-25}{90 \times 10^{-3}} \end{aligned}$$

$$\frac{dQ}{dt} = 77.78 \text{ W}$$

Solution

4. Given: $A = 1.5 \text{ m}^2$, $x = 0.2 \times 10^{-2} \text{ m}$,
 $T_i = 15^\circ\text{C}$, $T_f = 0^\circ\text{C}$, $k = 0.84 \text{ W m}^{-1} \text{ K}^{-1}$,

(a) $\frac{dQ}{dt} = ?$

$$\frac{dQ}{dt} = -kA \frac{T_f - T_i}{x}$$

$$= -0.84 \times 1.5 \times \frac{0 - 15}{0.2 \times 10^{-2}}$$

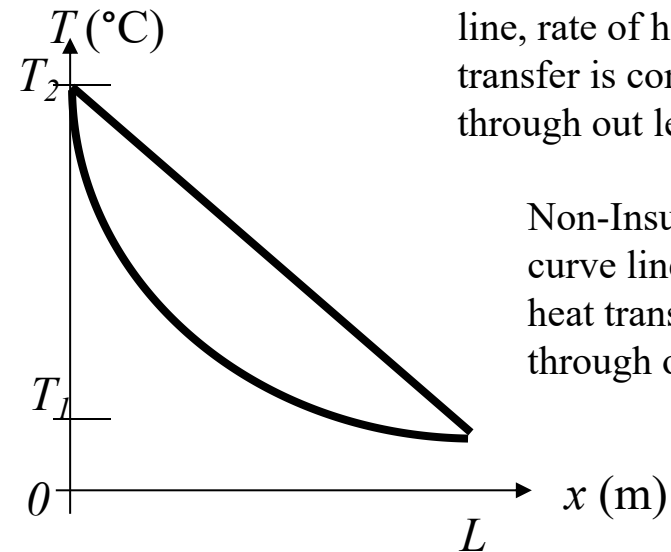
$$\frac{dQ}{dt} = 9450 \text{ W}$$

(b) To reduce heat flow;

- Increase thickness of window
- Reduce area window

Solution

5. Graph temperature gradient for insulated & non insulated rod



Insulated: straight line, rate of heat transfer is constant through out length

Non-Insulated: curve line, rate of heat transfer is not through out length



Learning Outcomes

8.2 IDEAL GAS EQUATION

At the end of this chapter, students should be able to:

- a) State** Gas's Law
- b) Sketch** the following graphs of an ideal gas :
 - i) p-V graph at constant temperature
 - ii) V-T graph at constant pressure
 - iii) p-T graph at constant volume
- c) Explain** the following graphs of an ideal gas :
 - i) p-V graph at constant temperature
 - ii) V-T graph at constant pressure
 - iii) p-T graph at constant volume
- d) State** ideal gas equation.
- e) Use** ideal gas equation, $pV = nRT$

8.2 The Gas Law

Boyle's Law

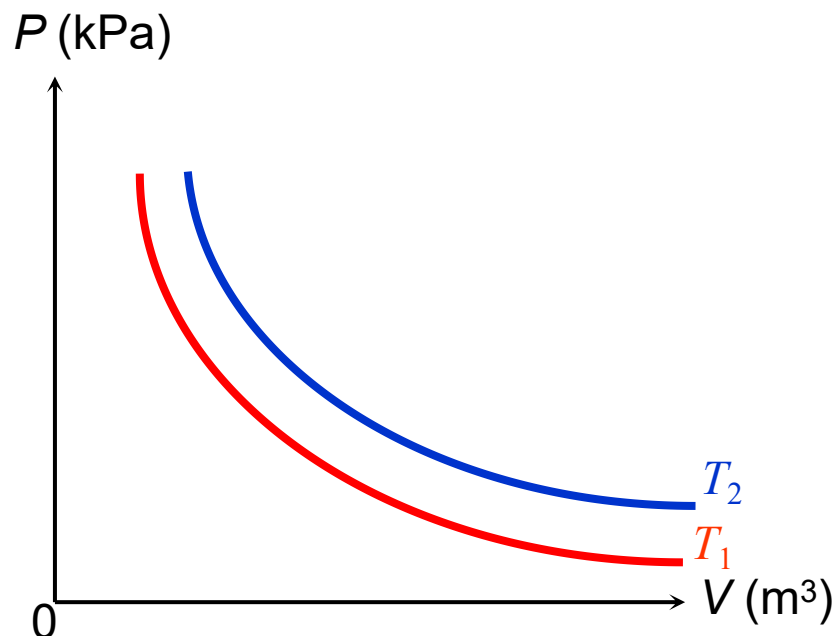
For a fixed amount of gas at a constant temperature, gas **pressure is inversely proportional** to gas **volume**.

$$P \propto \frac{1}{V}$$

$$PV = \text{constant}$$

$$P_1 V_1 = P_2 V_2$$

Graph of the Boyle's law



Charles's Law

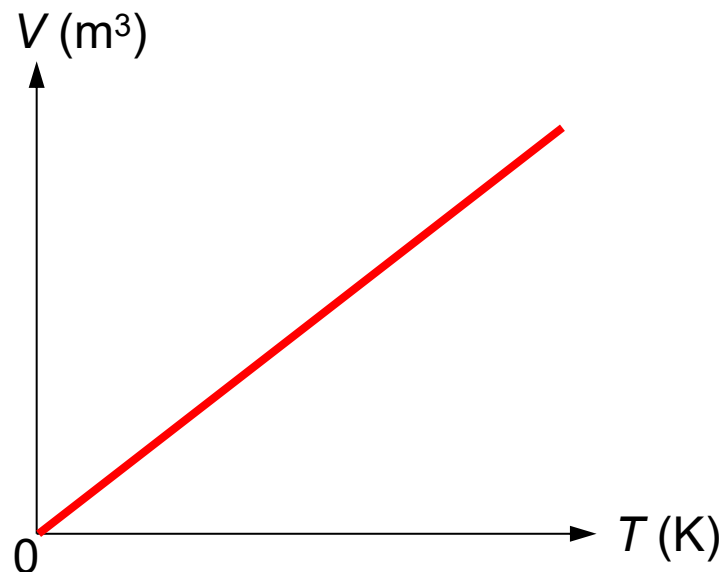
For a fixed amount of gas at a constant pressure, gas **volume** is **directly proportional** to its absolute **temperature**.

$$V \propto T$$

$$\frac{V}{T} = \text{constant}$$

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

Graph of the Charles's law



Gay-Lussac / Pressure Law

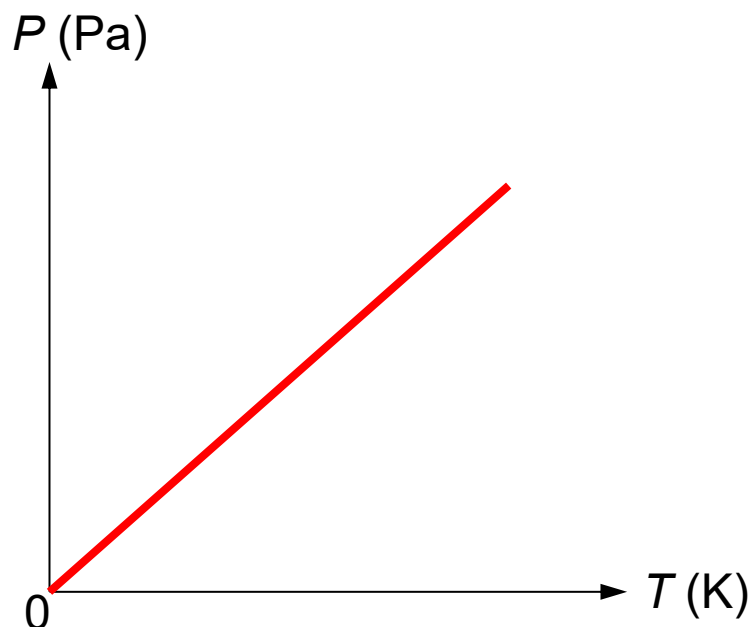
For a fixed amount of gas at a constant volume, gas **pressure** is **directly proportional** to its absolute **temperature**.

$$P \propto T$$

$$\frac{P}{T} = \text{constant}$$

$$\frac{P_1}{T_1} = \frac{P_2}{T_2}$$

Graph of the pressure law



8.2d The Ideal Gas Law

An ideal gas law obeys all the gas laws over all pressure and temperature

These 3 laws can be combined into a single more general relation between absolute **pressure**, **volume** and absolute **temperature** of a fixed quantity of gas:

Write in mathematics form: $\frac{PV}{T} = \text{constant}$

1 mole of any gas at standard temperature & pressure(STP),

$T = 273.15 \text{ K}$, $P = 101.3 \text{ kPa}$ occupies $V = 22.4 \text{ liters} = 22.4 \times 10^{-3} \text{ m}^3$:

Using
$$\frac{PV}{T} = \frac{101.3 \times 10^3 \times 0.0224}{273.15} = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$$

8.31 J K⁻¹ mol⁻¹ is known as
molar gas constant, R

value of **R** is
found
experimentally
same for all
gases.

In general, if n moles occupy a volume V at pressure P & absolute temperature T , we have :

$$PV = nRT \quad [\text{Ideal Gas Equation}]$$

□ The number of moles of gas can also be expressed as:

$$n = \frac{m}{M} \quad \text{or} \quad n = \frac{N}{N_A}$$

where

P = pressure of gas in Pascal

V = volume of gas in m³

T = temperature of gas in K

n = number of moles

R = gas constant 8.31 J K⁻¹ mol⁻¹

m = mass of gas in kg

M = molar mass of gas in kg mol⁻¹

N = number of gas molecules

N_A = Avogadro Number 6.02×10^{23} atoms

Relationship between **number of moles**, **number of particles**, **mass** and the **volume of gas**:

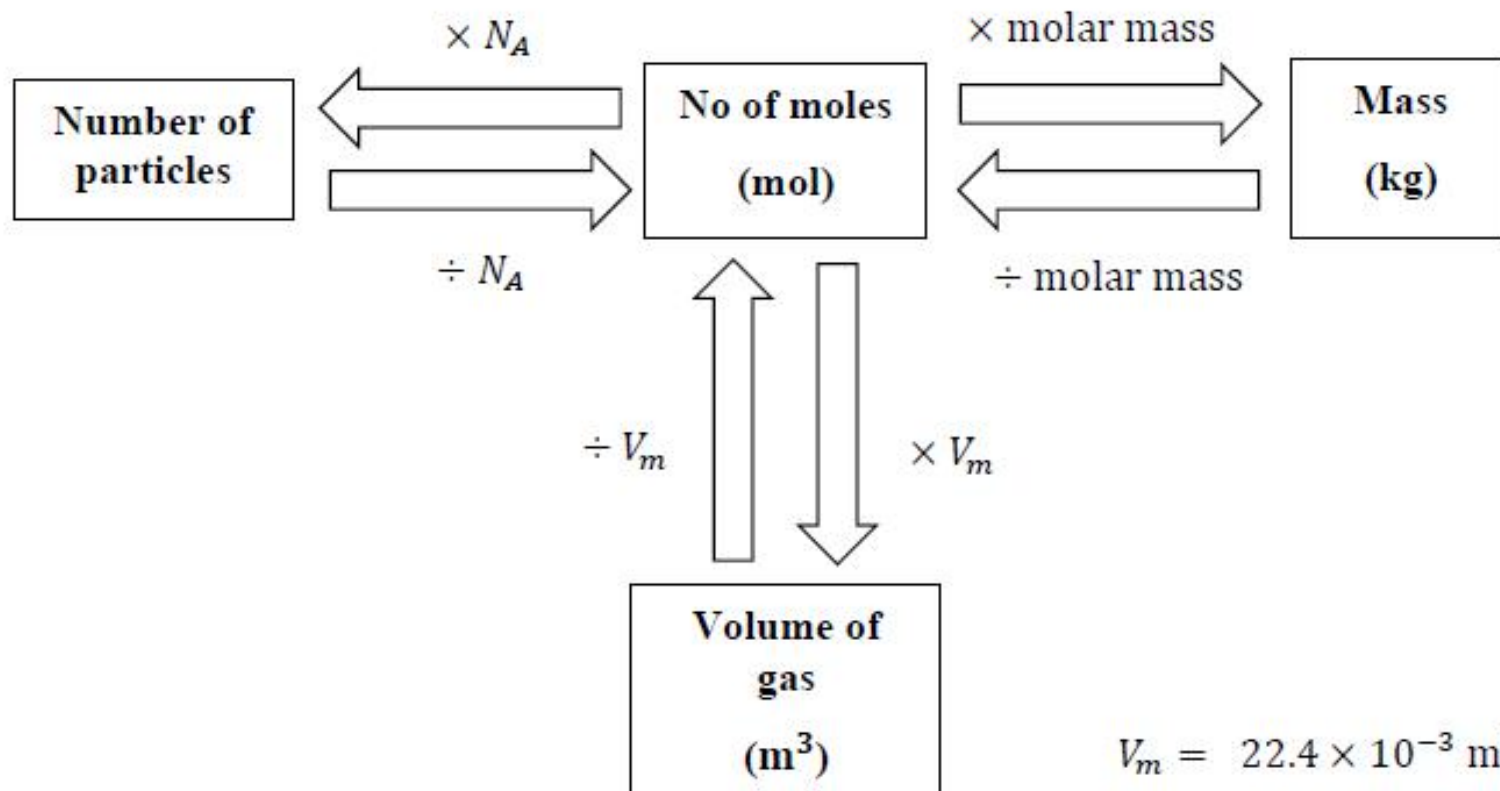


Figure 8-6 The mole is the central to all conversion between mass, volume and number of particles

Example 4

What is the approximate number of atoms in a cubic metre of an ideal monoatomic gas at a temperature of 27 °C and a pressure of 1×10^5 Pa ?

Solution

Given: $P = 1 \times 10^5$ Pa, $T = (27 + 273.15) = 300.15$ K

$V = 1 \text{ m}^3$ (cubic metre), $N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$

$R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$, number of atoms, $N = ?$

Using : $PV = \left(\frac{N}{N_A}\right)RT$

$$N = \frac{(1 \times 10^5)(1)(6.02 \times 10^{23})}{(8.31)(300.15)}$$

$$N = 2.41 \times 10^{25}$$

Exercise 8.2

1. A gas cylinder is fitted with a safety valve which releases gas when the pressure inside the cylinder reaches 2.0×10^6 Pa. Given that the maximum mass of this gas the cylinder can hold at 10°C is 15 kg, what would be the maximum mass at 30°C ?
2. A cylinder contains 12 L of oxygen at 20°C and 15 atm. The temperature is raised to 35°C and the volume is reduced to 8.5 L. What is the final pressure of the gas in atmosphere? Assume that the gas is ideal.
3. Estimate the number of molecules in a flask of volume $5.0 \times 10^{-4} \text{ m}^3$ which contains oxygen gas at a pressure of 2.0×10^5 Pa and temperature of 300 K.
4. A cylinder contains a hydrogen gas of volume $2.40 \times 10^{-3} \text{ m}^3$ at 17°C and 2.32×10^6 Pa. Calculate
 - a. the number of moles of hydrogen in the cylinder,
 - b. the number of molecules of hydrogen in the cylinder,
 - c. the mass of the hydrogen (given the molar mass of hydrogen = 2 g mol^{-1}).

Exercise 8.2

5. A flask with volume of 1.5 L, provided with a stop-cock contains ethane gas at 300 K and 1 atm pressure. The molar mass of the gas is 30.1 g mol⁻¹. The system is warmed to a temperature 490 K, with the stop-cock open to the atmosphere. The stop-cock is then closed, and the flask is cooled to its original temperature.

- a. What is the final pressure of the gas in the flask?
- b. How many grams of ethane gas remain in the flask?

ANS:

1. 14 kg

2. 22.3 atm

3. 2.41×10^{22} molecules

4. a) 2.31 mol b) 4.62 g

5. a) 6.2×10^4 Pa, b) 1.12 g

Solution

1. Given: $P = 2 \times 10^6 \text{ Pa}$, $m_1 = 15 \text{ kg}$,
 $T_1 = 10 + 273.15 = 283.15 \text{ K}$,
 $T_2 = 30 + 273.15 = 303.15 \text{ K}$, $m_2 = ?$

$$PV = \frac{mRT}{M_r} \quad \rightarrow m \text{ is inversely proportional to } T$$

$$m_2 = \frac{T_1}{T_2} m_1$$

$$= \frac{283.15}{303.15} 15$$

$$m_2 = 14.0 \text{ kg}$$

Solution

2. Given: $V_1 = 12 \text{ L}$, $T_1 = 20 + 273.15 = 293.15 \text{ K}$,
 $P_1 = 15 \text{ atm}$, $V_2 = 8.5 \text{ L}$,
 $T_2 = 35 + 273.15 = 308.15 \text{ K}$,
 $P_2 = ?$

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

$$P_2 = \frac{P_1 V_1}{T_1} \frac{T_2}{V_2}$$

$$= \frac{15 \times 12}{293.15} \times \frac{308.15}{8.5}$$

$$P_2 = 22.3 \text{ atm}$$

Solution

3. Given: $V = 5 \times 10^{-4} \text{ m}^3$, $P = 2 \times 10^5 \text{ Pa}$, $T = 300 \text{ K}$, $N = ?$

$$PV = \frac{NRT}{N_A}$$

$$N = \frac{PV N_A}{RT}$$

$$= \frac{2 \times 10^5 \times 5 \times 10^{-4} \times 6.02 \times 10^{23}}{8.31 \times 300}$$

$$N = 2.41 \times 10^{22} \text{ molecules}$$

Solution

4. Given: $V = 2.4 \times 10^{-3} \text{ m}^3$, $P = 2.32 \times 10^6 \text{ Pa}$, $T = 17 + 273.15 = 290.15 \text{ K}$, $M_r = 2 \text{ g mol}^{-1}$

a. $n = ?$

$$PV = nRT$$

$$n = \frac{PV}{RT}$$

$$= \frac{2.32 \times 10^6 \times 2.4 \times 10^{-3}}{8.31 \times 290.15}$$

$$n = 2.31 \text{ mol}$$

b. $N = ?$

$$n = \frac{N}{N_A}$$

$$= 2.31 \times 6.02 \times 10^{23}$$

$$N = 1.39 \times 10^{24} \text{ molecules}$$

c. $m = ?$

$$n = \frac{m}{M_r}$$

$$= 2.31 \times 2$$

$$m = 4.62 \text{ g}$$

Solution

5. Given: $V = 1.5 \text{ L} = 1.5 \times 10^{-3} \text{ m}^3$, $P_1 = 1.01 \times 10^5 \text{ Pa}$, $T_1 = 490 \text{ K}$, $M_r = 30.1 \text{ g mol}^{-1}$
 $T_2 = 300 \text{ K}$

a. $P_2 = ?$

$$\frac{P_1}{P_2} = \frac{T_1}{T_2}$$

$$P_2 = \frac{T_2}{T_1} P_1$$
$$= \frac{300}{490} \times 1.01 \times 10^5$$

$$P_2 = 6.2 \times 10^4 \text{ Pa}$$

b. $m = ?$

$$PV = \frac{mRT}{M_r}$$

$$m = \frac{PVM_r}{RT}$$

$$= \frac{1.01 \times 10^5 \times 1.5 \times 10^{-3} \times 30.1 \times 10^{-3}}{8.31 \times 490}$$

$$m = 1.12 \times 10^{-3} \text{ kg}$$



Learning Outcomes

8.3 THERMODYNAMICS

At the end of this chapter, students should be able to:

- a) State** the first law of thermodynamics.
- b) Solve** problems related to the first law of thermodynamics.
- c) Define** the thermodynamics processes :
 - i) Isothermal
 - ii) Isochoric
 - iii) Isobaric
 - iv) Adiabatic
- d) Interpret** p - V graph for all the thermodynamics processes.
(Experiment 6 : Heat)



What is **THERMODYNAMICS???**

“is the branch of physical science that deals with the relations between heat and other forms of energy (such as mechanical, electrical, or chemical energy), and, by extension, of the relationships between all forms of energy.”

There are 4 laws of thermodynamics ...



8.3a First Law of Thermodynamics

State : “ The heat (Q) supplied to a system **is equal** to the **increase in the internal energy** (ΔU) of the system **plus** the **work done** (W) **by the system** on its surroundings”

$$Q = \Delta U + W$$

where :

Q : quantity of heat supplied

W : work done

ΔU : change in internal energy ($U_{\text{final}} - U_{\text{initial}}$)

8.3b Solve Problem Thermodynamics

Change in Internal energy, ΔU

Suppose gas starts from state A, undergoes thermodynamic changes & attains a final state B.

$$\Delta U_{AB} = U_{final(B)} - U_{initial(A)}$$

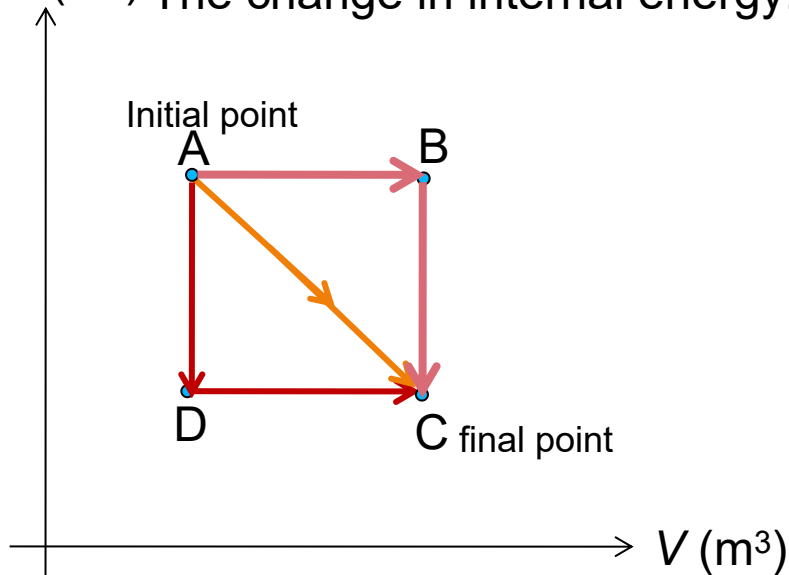
ΔU independent of the path taken but dependent only on the initial & final states of the gas.

$$\Delta U_{AC} = \Delta U_{ABC} = \Delta U_{ADC}$$

If the process is reverse as in **C** becomes the **initial point** and **A** is the **final**,

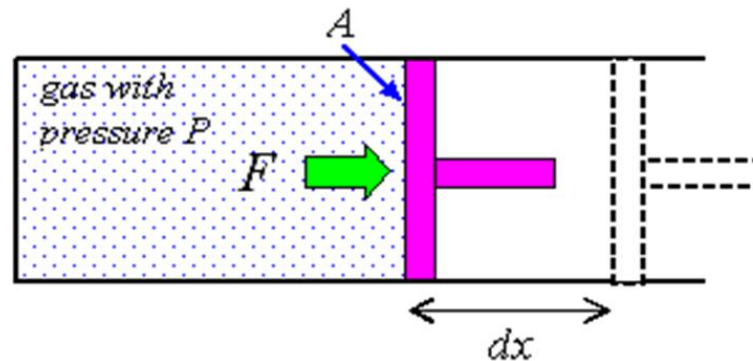
$$\Delta U_{CA} = - \Delta U_{AC}$$

P (Pa) The change in internal energy:

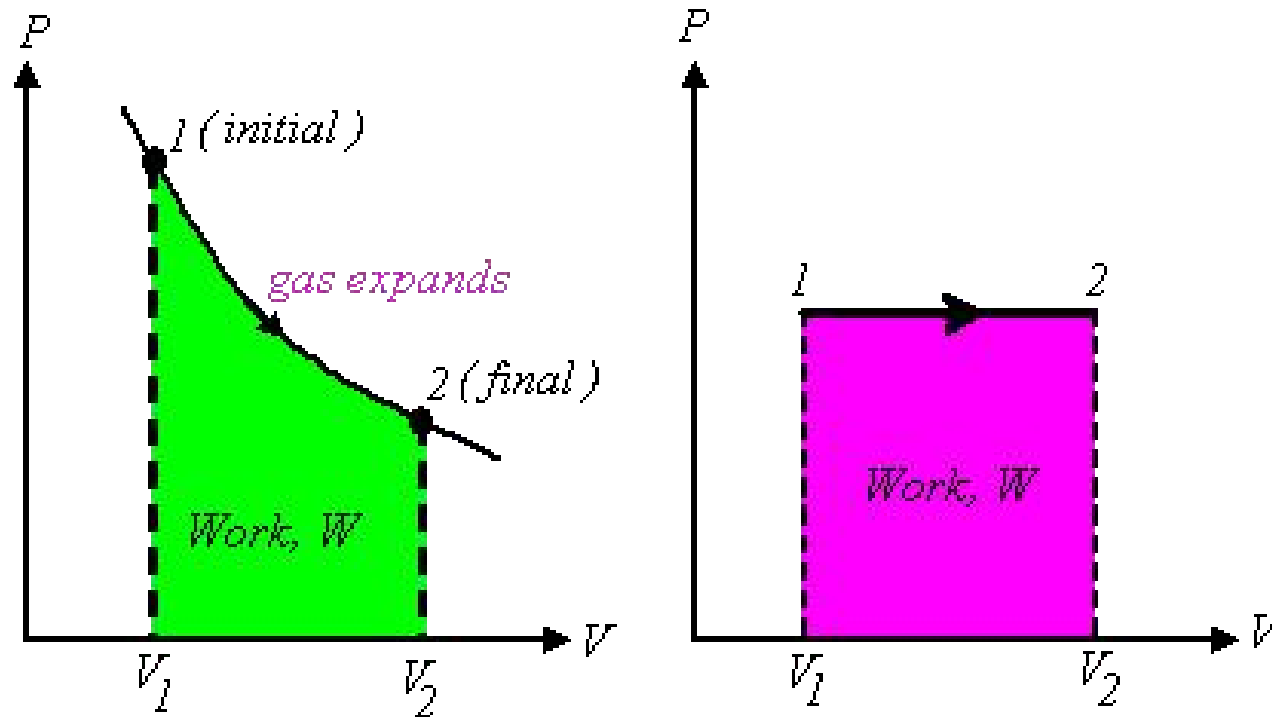


The work dW done on / by the gas (system) is determined by sign convention:

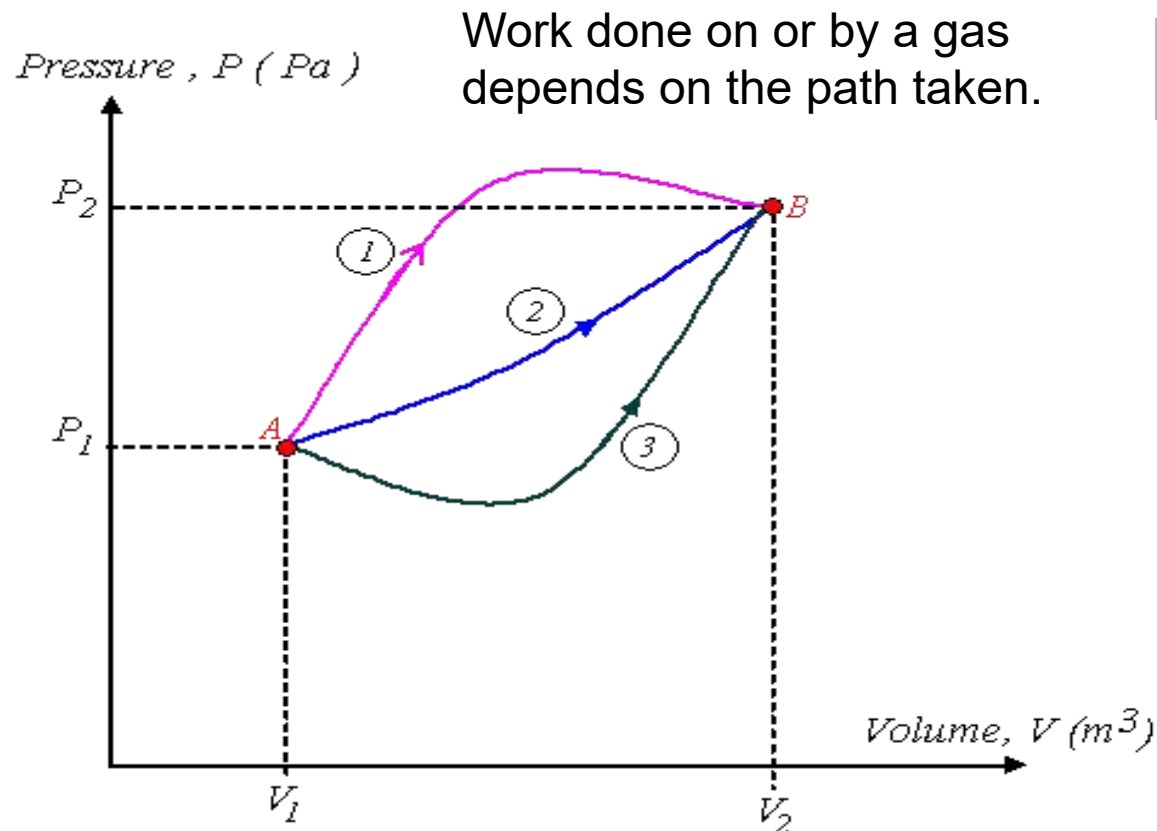
$+ W$	$- W$
Gas expansion	Gas compression
$V_2 > V_1$ size container gets bigger	$V_2 < V_1$ size container gets smaller
Work is done by the gas/system	Work is done on the gas/system



Work done from $P - V$ Graphs



Work, W = Area under the process curve



$$W_1 \neq W_2 \neq W_3$$

Why ?
Because area
under the curve
for path 1 $\neq$
path 2 $\neq$ path 3

But.. $\Delta U_1 = \Delta U_2 = \Delta U_3$

ΔU only depends on initial and final point

Sign convention of Q , ΔU , W

	Positive value	Negative value
Q	System absorb heat / Heat flow into the system	System loses heat / Heat flow out of the system
ΔU	U increases ($T \uparrow$)	U decreases ($T \downarrow$)
W	Work done by the system	Work done on the system

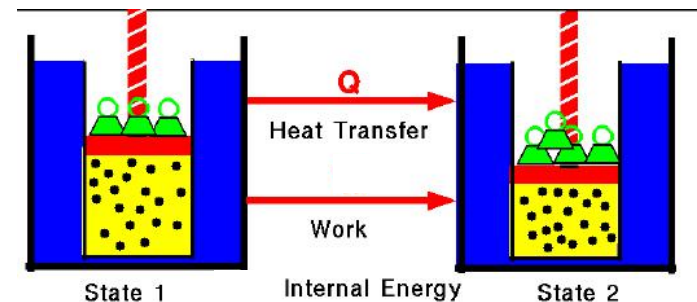


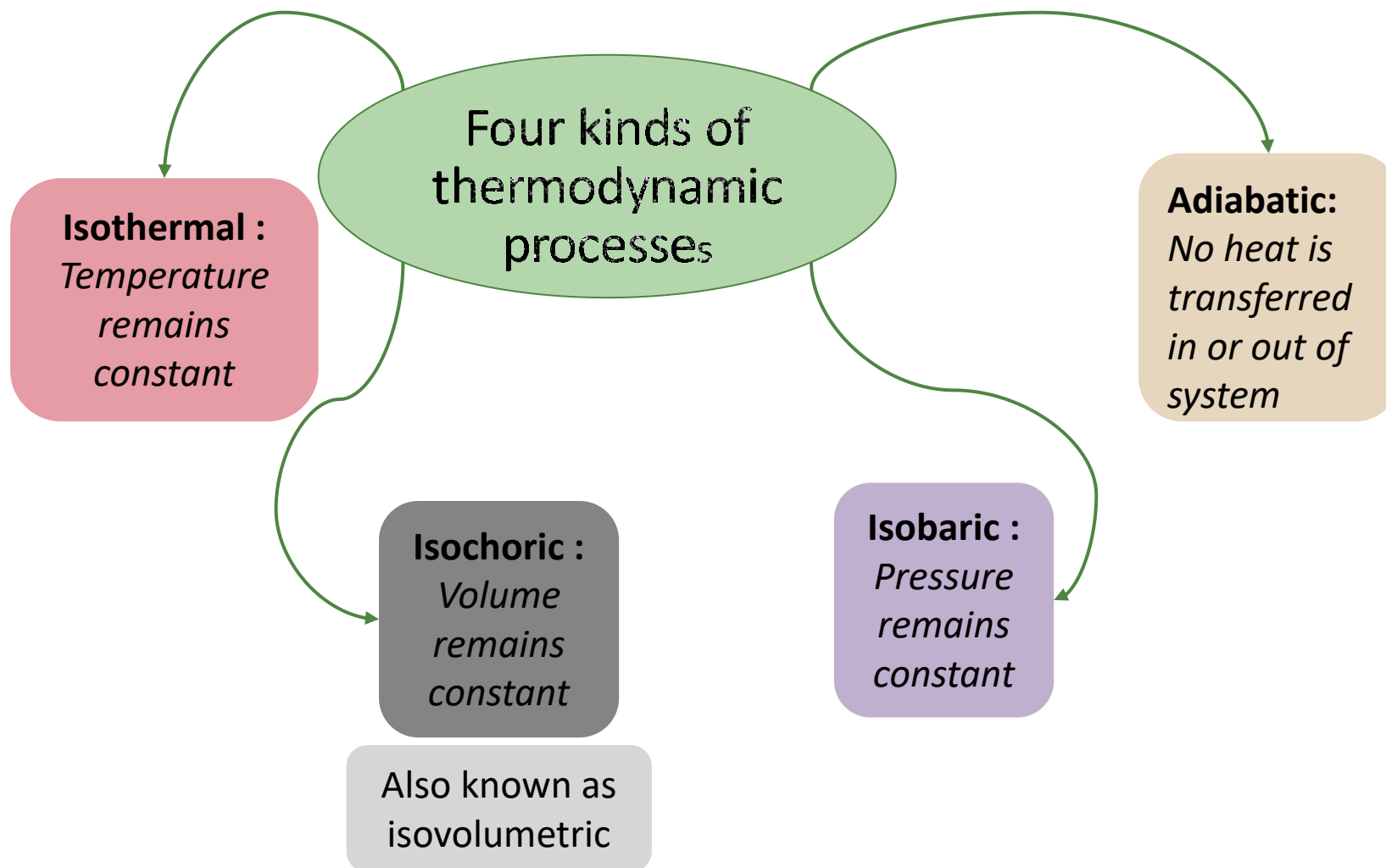
8.3c Thermodynamic Processes

A thermodynamic process is process where the parameters of a system such as pressure, volume & temperature undergo changes.

This can happen due to either :

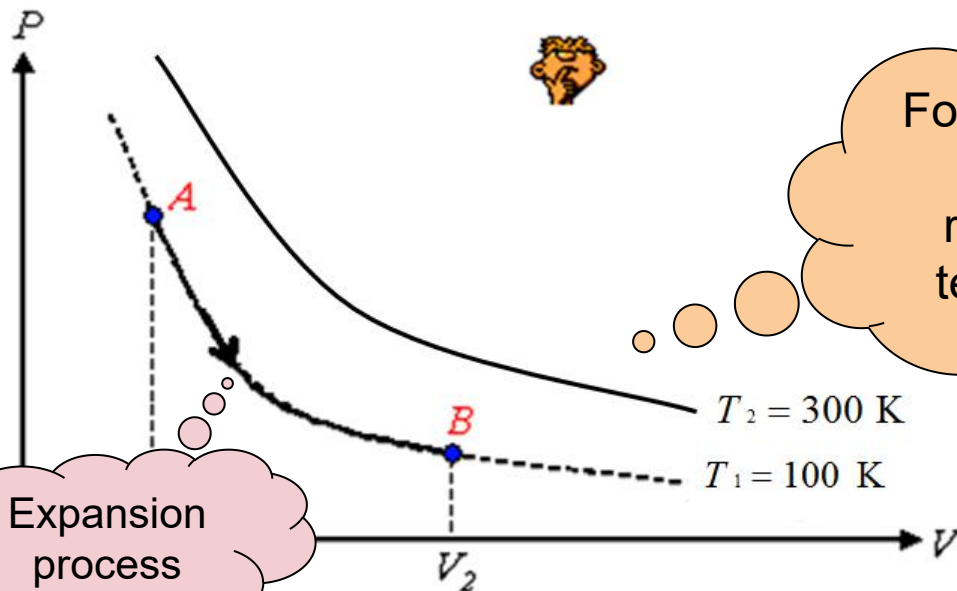
- (a) Exchange of heat between system & surrounding.
- (b) Mechanical work being done on or by the system (ex: gas expand)
- (c) Both of (a) & (b)





1. Isothermal Process

Occur at constant temperature



P & V both vary!
 U constant (T constant)
 $\Delta U = 0$
 $Q = W$ (+ve)

Isothermal
process
obey
Boyle's
Law



$$W = nRT \ln \left(\frac{V_2}{V_1} \right)$$

Slow
process

$$P_1 V_1 = P_2 V_2$$

Obey Boyle's law

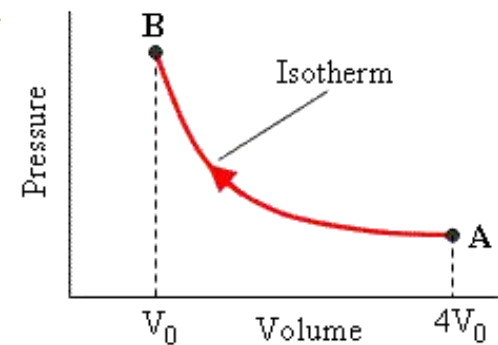
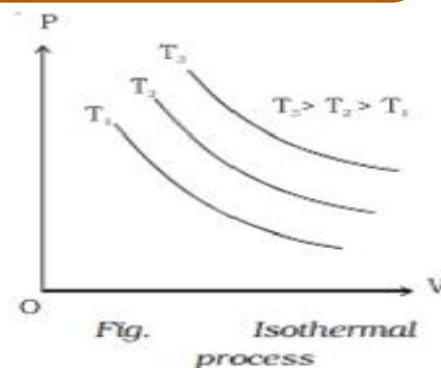
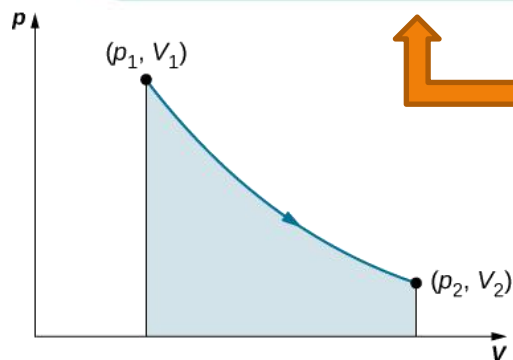
Gas expand, $W > 0$

Isothermal
(T constant)

Gas compressed,
 $W < 0$

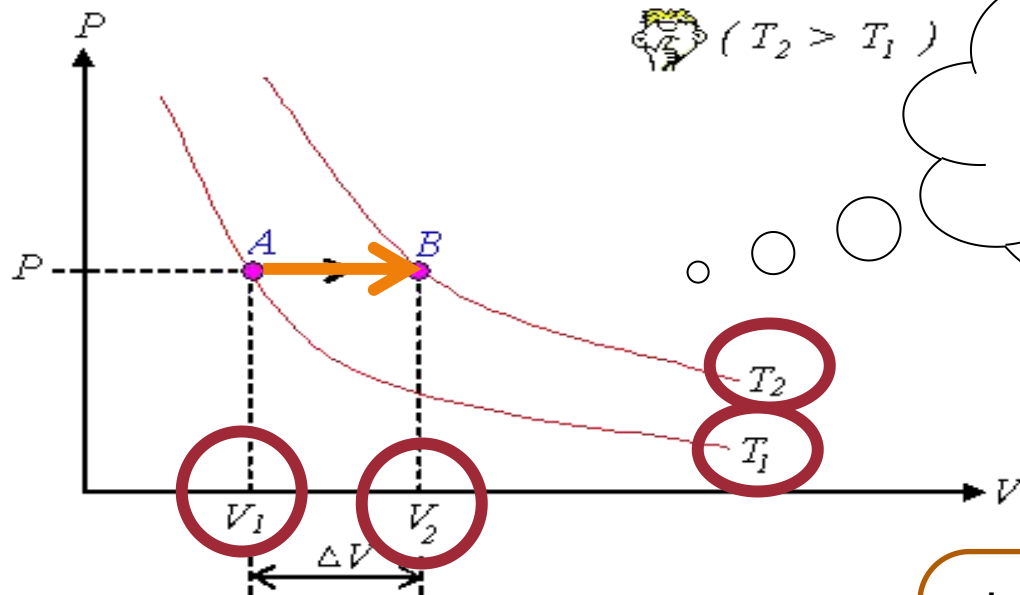
$$Q = W$$

heat added to the gas
goes into doing work



2. Isobaric Process

Occur at constant pressure



When heated, gas expands
Gas crosses to higher temperature

$T \uparrow \rightarrow U \uparrow \rightarrow \Delta U > 0$ (+ve)
 $\rightarrow V$ changes
 $\rightarrow W \neq 0$

Isobaric process obey Charles's Law



$$W = P(V_2 - V_1) = P\Delta V$$

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

Obey Charles' law

Gas expand,
 $W > 0$

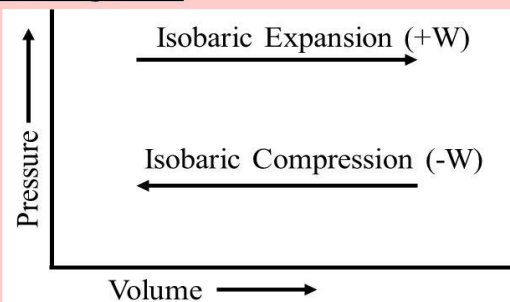
Isobaric
(P constant)

Gas compressed,
 $W < 0$

$$Q = \Delta U + P(V_2 - V_1)$$

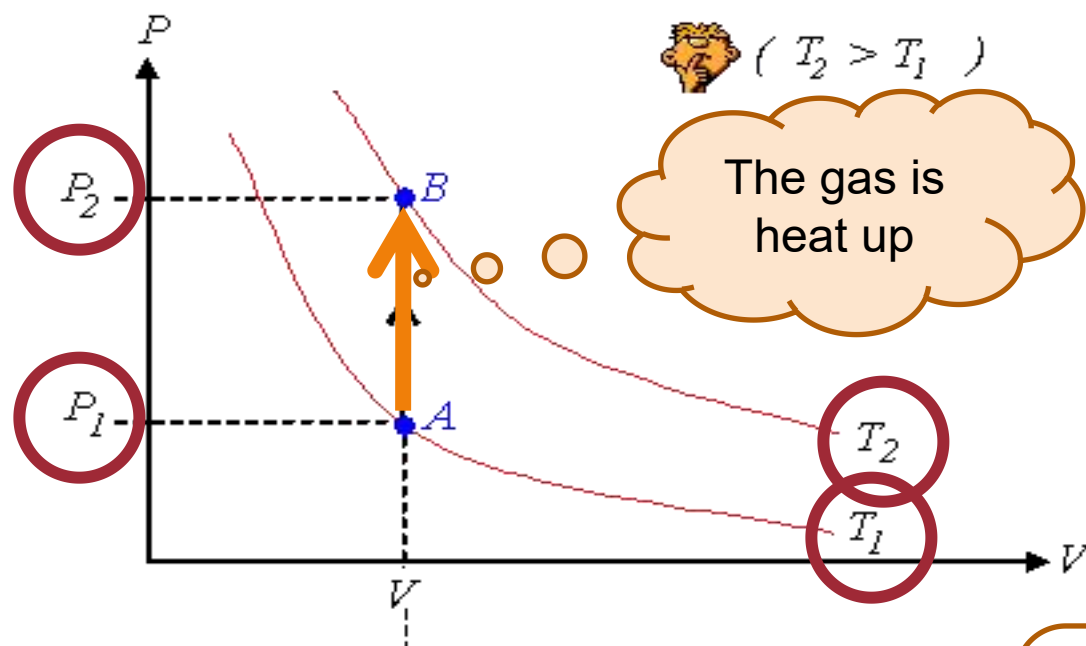
heat added to the gas goes into doing work and increase its internal energy

P-V diagrams



3. Isochoric Process

Occur at constant volume



No work is done since the area under such a curve is zero.
(No change in volume) $W = 0$

Isochoric process obey Pressure Law



$$\frac{P_1}{T_1} = \frac{P_2}{T_2}$$

Obey Pressure law

**Gas heated,
T increase
W = 0**

**ISOCHORIC
(V constant)**

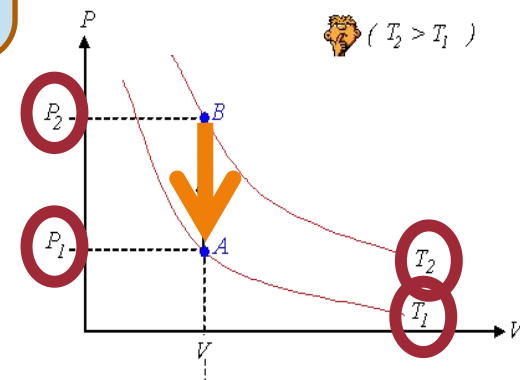
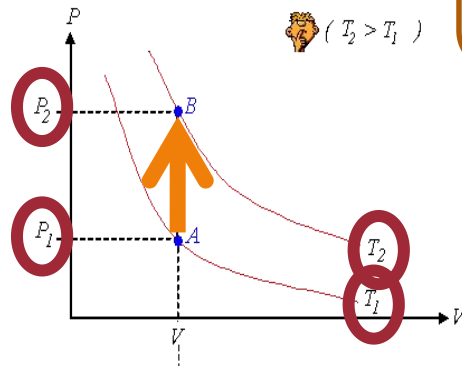
**Gas cooled,
T decrease
W = 0**

**Heat added to
increase its
internal energy**

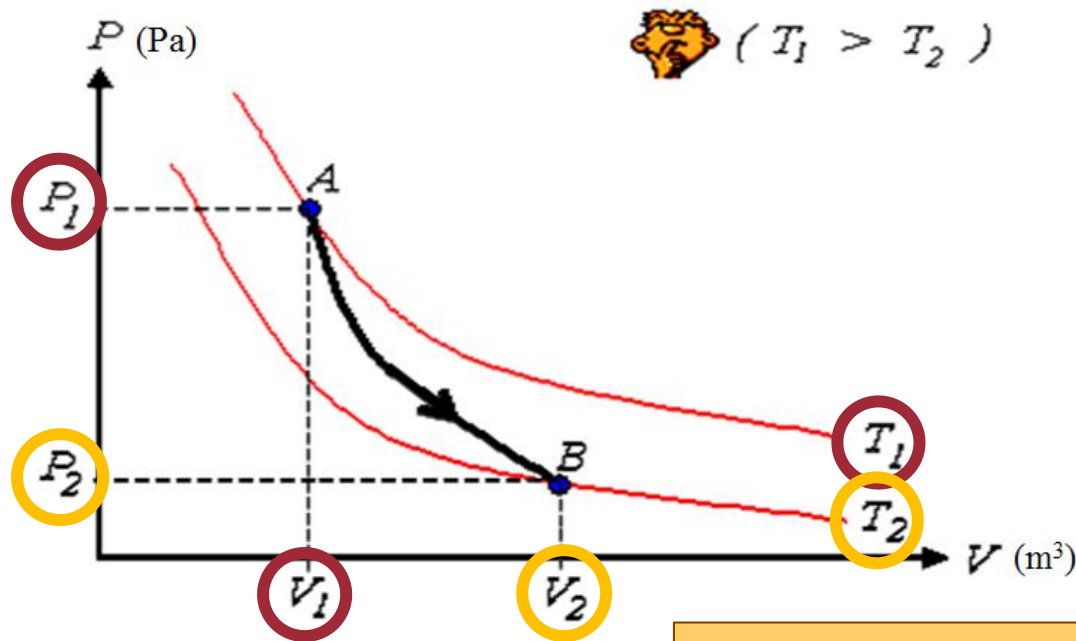
**Heat removed to
decrease its
internal energy**

$$Q = \Delta U + W$$

$$Q = \Delta U$$



4. Adiabatic Process - No heat transferred in or out of gas, so $Q = 0$.



all quantities changes!

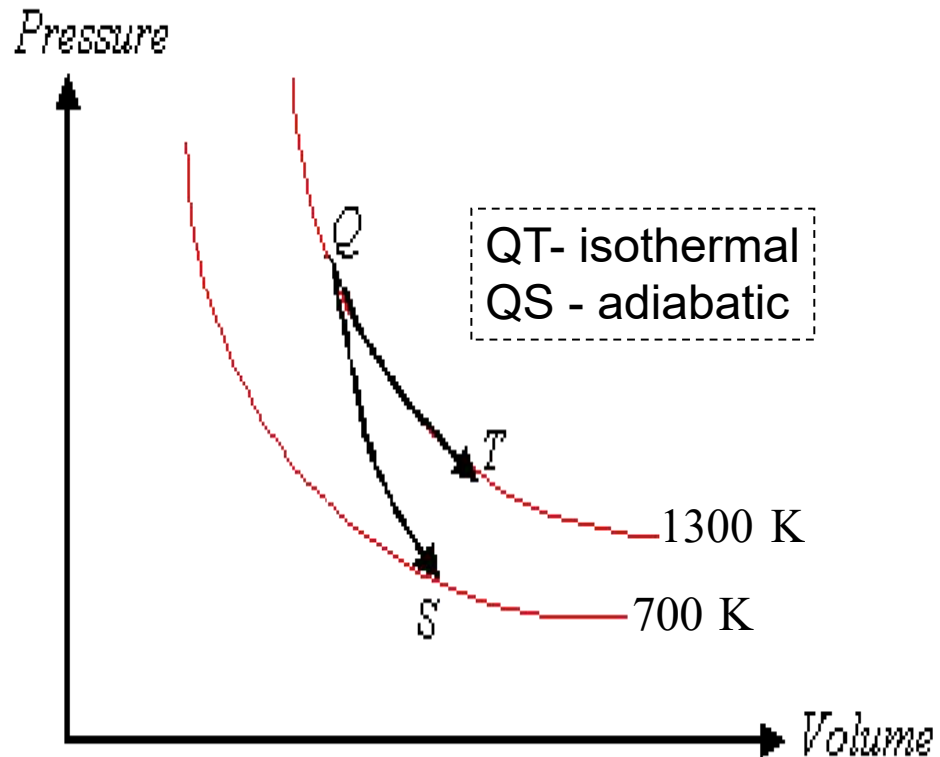
From A to B, P & $T \downarrow$ but $V \uparrow$
Such process is called cooling down

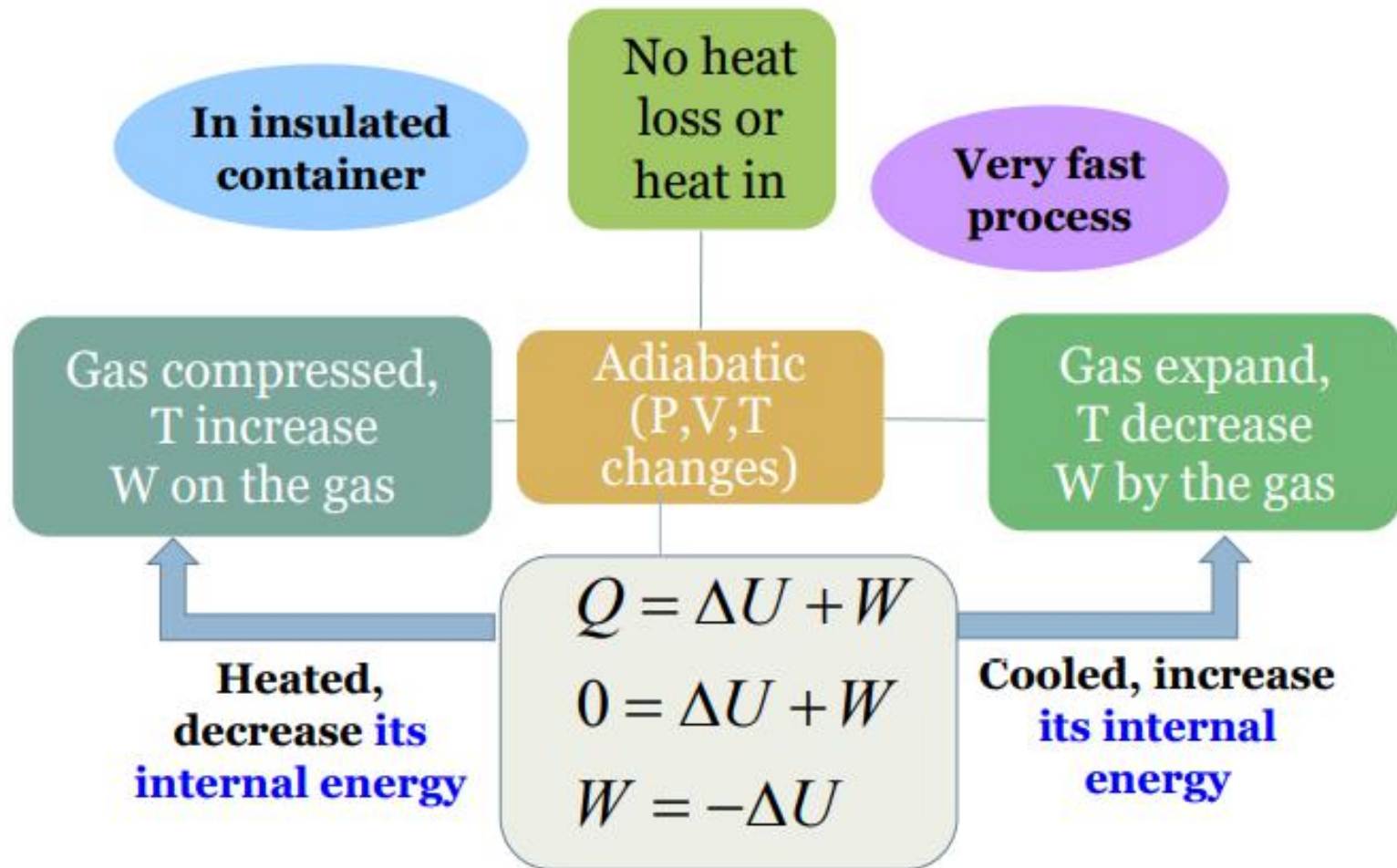
- As gas expands, it does a +ve work
- Internal energy decreases (-ve)
- Temperature drops
- Note that an adiabatic curve at any point is always steeper than an isothermal at that point!

Adiabatic change DOES NOT obey Boyle's law.
The gas temperature is NO longer constant.

Adiabatic curve is **steeper** than isothermal curve because in **adiabatic there is a changes in internal energy**

(temperature change)
while in **isothermal internal energy, U remain unchanged**
(temperature constant).





Example 5

8000 J of heat is removed from a refrigerator by a compressor which has done 5000 J of work. What is the change in the internal energy of the gas in the system?

Solution

Given: $Q = -8000$ J (gas removed : -ve value), $W = -5000$ J,
 $\Delta U = ?$

$$Q = W + \Delta U$$

$$\Delta U = -8000 - (-5000)$$

$$\Delta U = -3000 \text{ J}$$

Example 6

An ideal gas absorbs 1.5 kJ of heat & expands from its initial volume of 20 litres to 35 litres under constant pressure of 30 kPa. Determine the change in internal energy of the gas.

Solution

Given: $Q = +1.5 \text{ kJ} = +1.5 \times 10^3 \text{ J}$ (gas absorbs 1.5 kJ of heat : +ve value)

$$V_1 = 20 \text{ L} = 20 \times 10^{-3} \text{ m}^3, V_2 = 35 \text{ L} = 35 \times 10^{-3} \text{ m}^3$$

$$P = 30 \times 10^3 \text{ Pa}, \Delta U = ?$$

Apply 1st law of thermodynamic : $Q = \Delta U + W$, but need to find the value of W first



Using :

$$W = P(V_2 - V_1)$$

$$= 30 \times 10^3 (35 \times 10^{-3} - 20 \times 10^{-3})$$

$$W = -450 \text{ J}$$

Using 1st Law of Thermodynamics :

$$Q = \Delta U + W$$

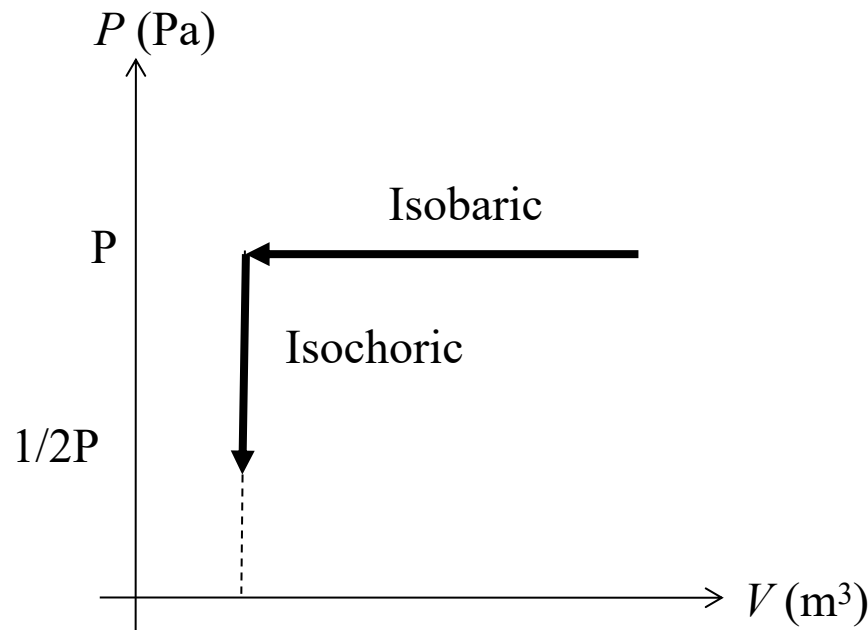
$$= 1500 - 450$$

$$\Delta U = 1050 \text{ J}$$

* U of the system increases

Example 7

A gas is compressed under constant pressure, P and then cooled down until the pressure is half P . Sketch the pressure volume graph and labelled the processes.

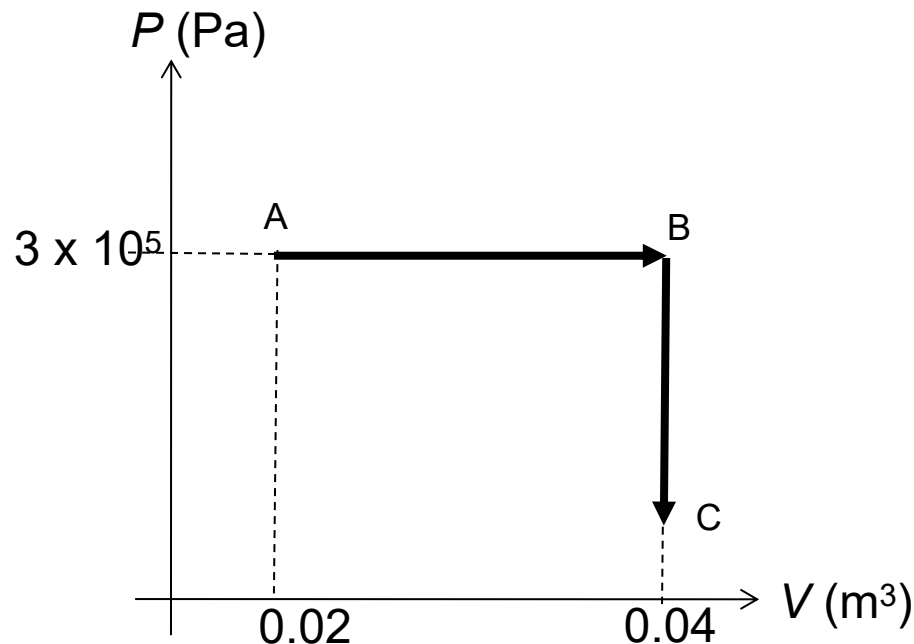
Solution

Exercise 8.3

1. An ideal gas at an initial volume 1.5 L undergoes isobaric process at a constant pressure 1 atm to 3 L. It is then undergoing cooling process at constant volume until pressure is 0.5 atm. The gas is finally compressed isothermally back to its initial state. Sketch P-V graph shows the processes involved.
2. A fixed mass of ideal gas is contained in a cylinder, the cylinder volume can be varied by moving a piston in or out. The gas has an initial volume 0.01 m^3 at 100 kPa pressure and temperature 400 K. The gas is cooled at constant pressure until its volume is 0.005 m^3 . Sketch the P-V graph for the process.
3. A system absorbs 2090 J of heat and at the same time does 400 J of work. What is the change in the internal energy of the system?

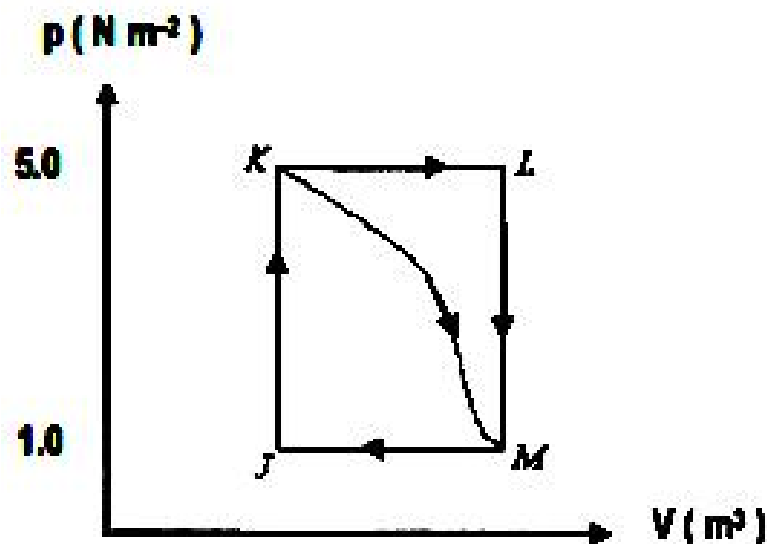
Exercise 8.3

4. A gas has undergoes two processes AB and BC. 8 kJ of heat is supplied to the process AB and 4 kJ of heat is removed from the gas in process BC.
- Calculate the work done for AB and BC and
 - the change in internal energy for AB and BC.



Exercise 8.3

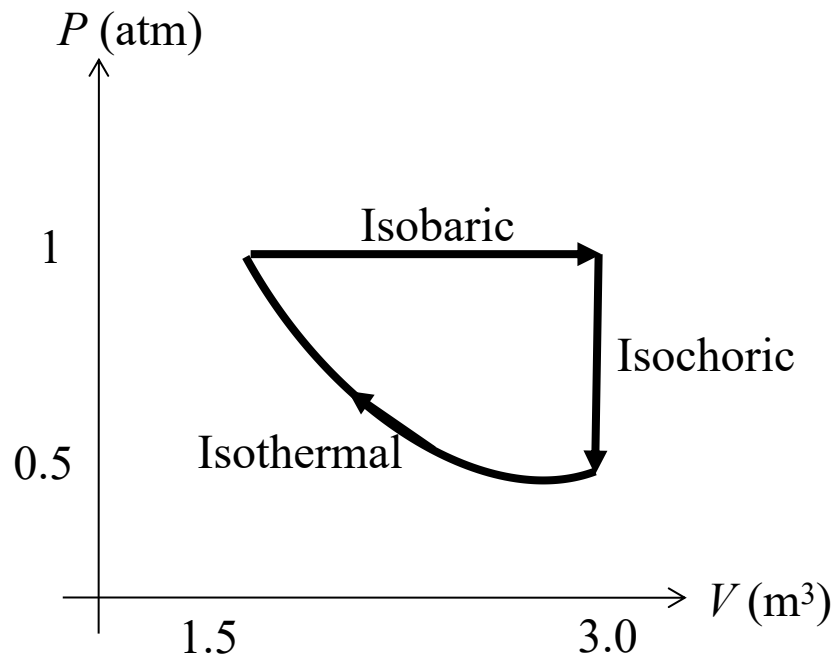
5. Figure shows a P-V diagram of an ideal gas. Change in internal energy from K to M is + 400 J, while the work done when the gas undergoes a change following a path KLM is +100 J.
- Calculate the heat flows into the system that follow the path of KLM.
 - Determine the work done by the system for process from M→J.
 - Calculate the change of heat with surrounding during process from M→K.



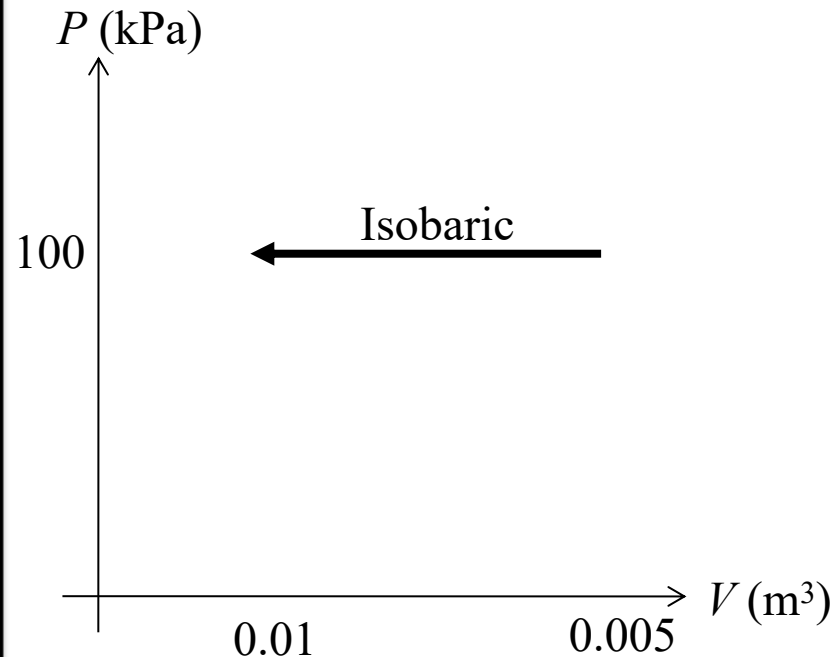
ANS:
 3. -1690 J
 4. a) 6000 J, 0 J, b) 2000 J, - 4000 J
 5. a) 500 J, b) - 20 m³, c) - 420 J

Solution

1. Given:

1st process: Isobaric expand2nd process: Isochoric cooling3rd process: Isothermal compressedSolution

2. Given: process isobaric compressed



Solution

3. Given: $Q = 2090 \text{ J}$ (absorb),
 $W = 400 \text{ J}$ (by system), $\Delta U = ?$

$$Q = W + \Delta U$$

$$\Delta U = 2090 - 400$$

$$\Delta U = -1690 \text{ J}$$

Solution

4. $Q_{AB} = 8000 \text{ J}$ (supplied), $Q_{BC} = -4000 \text{ J}$ (removed)

a) $W_{AB} = ?$ & $W_{BC} = ?$

$$W_{AB} = \text{area graph}$$

$$= 3 \times 10^5 (0.04 - 0.02)$$

$$W_{AB} = 6000 \text{ J}$$

$$W_{CD} = 0 \text{ J}$$

Isochoric
process

b) $\Delta U_{AB} = ?$ & $\Delta U_{BC} = ?$

$$Q_{AB} = W_{AB} + \Delta U_{AB}$$

$$\Delta U = 8000 - 6000$$

$$\Delta U_{AB} = 2000 \text{ J}$$

$$Q_{CD} = W_{CD} + \Delta U_{CD}$$

$$\Delta U = -4000 - 0$$

$$\Delta U_{CD} = -4000 \text{ J}$$

Solution

5. Given: $\Delta U_{KLM} = \Delta U_{KM} = +400 \text{ J}$, $W_{KLM} = W_{KL} + W_{LM} = 100 \text{ J}$, $W_{LM} = 0 \text{ J}$ (V const)

a) $Q_{KLM} = ?$

Using :
$$Q_{KLM} = W_{KLM} + \Delta U_{KLM}$$

$$= 100 + 400$$

$$Q_{KLM} = 500 \text{ J}$$

b) $W_{MJ} = ?$

Using : $W_{KL} = P\Delta V_{KL}$

$$\Delta V_{KL} = \frac{W_{KL}}{P}$$

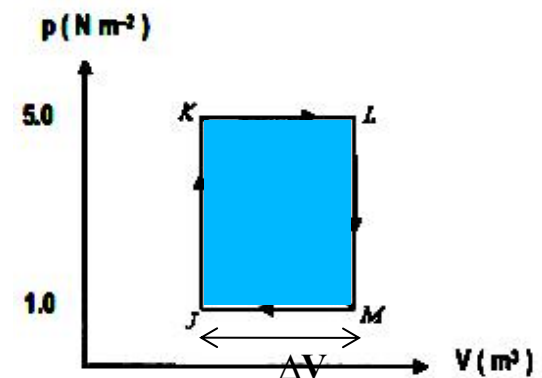
$$= \frac{100}{5} = 20 \text{ m}^3$$

$$\Delta V_{MJ} = -20 \text{ m}^3$$

$$W_{MJ} = P\Delta V_{MJ}$$

$$= 1(-20)$$

$$W_{MJ} = -20 \text{ J}$$



c) $Q_{\text{MJK}} = ?$

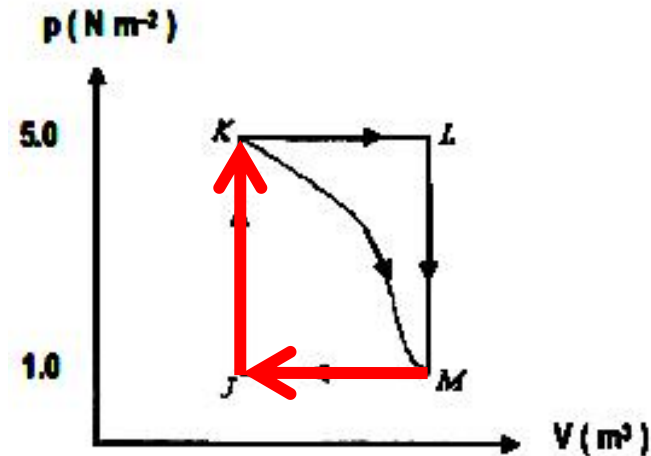
Knowing that $\Delta U_{\text{MK}} = -\Delta U_{\text{KM}} = -400\text{J}$

And $W_{\text{MJK}} = W_{\text{MJ}} + W_{\text{JK}} = W_{\text{MJ}} = -20\text{J}$

Where $W_{\text{JK}} = 0\text{J}$ because volume is constant.

Using: $Q_{\text{MJK}} = W_{\text{MJK}} + \Delta U_{\text{MJK}}$
 $= -20 - 400$

$Q_{\text{MJK}} = -420\text{ J}$



End of Chapter 8

“VERY BEST OF LUCK IN FINAL EXAM”

**「 NEVER GIVE UP ON YOUR DREAM;
ALWAYS REMAIN YOURSELF THE
REASON YOU ARE ON THIS JOURNEY 」**

**AS LONG AS YOU WILLING TO GIVE
YOUR VERY BEST, NOTHING IS
UNACHIEVEABLE !!!**